Experiment 2: Simulation and Comparison of Packet Switching and Circuit Switching

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Aim

To design and simulate packet switching and circuit switching using Cisco Packet Tracer and compare their performance and efficiency

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Theory

Packet Switching:
In packet switching, data is divided into small packets and transmitted independently across the network. Each packet may take a different path to reach the destination.

Circuit Switching:
In circuit switching, a dedicated communication path is established between sender and receiver before data transmission begins. The path remains fixed throughout communication.

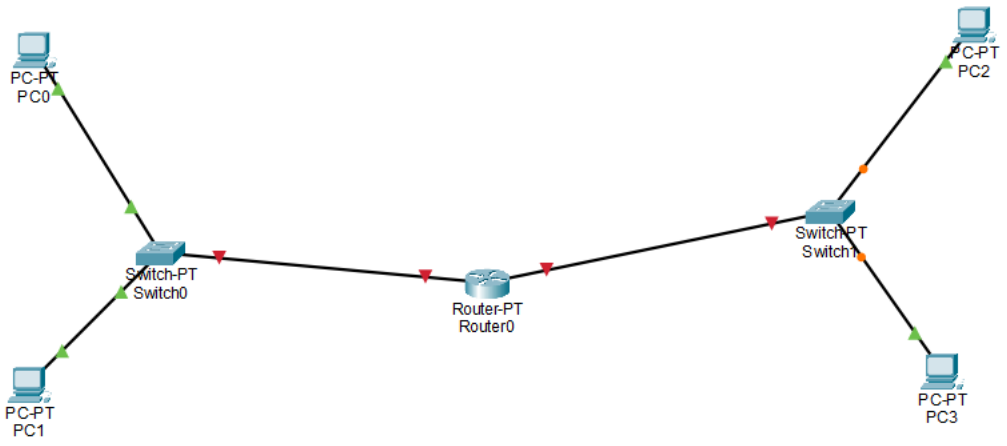
Requirements

- Cisco Packet Tracer
- PCs, Switches, Router
- Copper Straight-Through cables

PART A: PACKET SWITCHING

Step 1: Network Topology Design

- Place 4 PCs, 2 Switches, and 1 Router
- Connect devices as follows:
    - PC0, PC1 → Switch0
    - PC2, PC3 → Switch1
    - Switch0 → Router
    - Router → Switch1



Step 2: IP Configuration

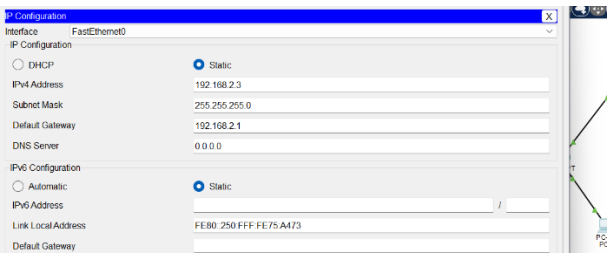
Assign IP addresses:

Left Network:

- PC0 → 192.168.1.2
- PC1 → 192.168.1.3
- Gateway → 192.168.1.1

Right Network:

- PC2 → 192.168.2.2
- PC3 → 192.168.2.3
- Gateway → 192.168.2.1



Step 3: Router Configuration

Open Router CLI and enter:

enable

configure terminal

interface fastEthernet 0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

interface fastEthernet 0/1

ip address 192.168.2.1 255.255.255.0

no shutdown

end

```
Press RETURN to get started!


Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastEthernet 0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed sta

Router(config-if)#inetface fastEthernet 0/0
                  ^
% Invalid input detected at '^' marker.

Router(config-if)#interface fastEthernet 0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Step 4: Connectivity Test

From PC0, run:

ping 192.168.2.2

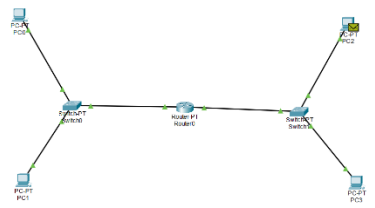
Verify successful replies.

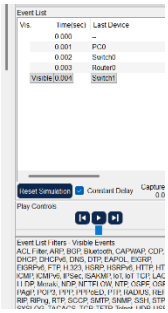
Step 5: Simulation Mode

- Switch to Simulation Mode
- Use “Add Simple PDU”
- Send packet from PC0 to PC2

Observe packet movement through network.

📸 *[Insert Screenshot: Packet Flow in Simulation Mode]*





Observations (Packet Switching)

- Data is transmitted in the form of packets
- Packets travel through multiple devices (switches and router)
- No fixed path is established
- Efficient use of network resources
- Flexible communication

PART B: CIRCUIT SWITCHING (SIMULATED)

Step 1: Network Setup

- Use existing topology or add 2 new PCs
- Ensure both devices are connected properly

Step 2: Continuous Communication

On PC, run:

ping 192.168.2.3 -t

This simulates continuous communication.

Step 3: Observe Communication

- Data flows continuously
- Same path is followed repeatedly
- No interruption in communication

Observations (Circuit Switching)

- Communication is continuous
- Path remains fixed during transmission
- Simulates dedicated connection
- Reliable communication
- Less efficient compared to packet switching

Comparison Between Packet Switching and Circuit Switching

Feature	Packet Switching	Circuit Switching
Path	Dynamic	Fixed
Efficiency	High	Low
Setup Time	Not required	Required
Flexibility	High	Low
Reliability	Moderate	High

Result / Conclusion

Packet switching is more efficient and flexible as it allows dynamic routing of data packets. Circuit switching provides a dedicated path ensuring stable communication but is less efficient. The simulation demonstrates the advantages and limitations of both techniques in data transmission.